

TESTING OF HYPOTHESES

Sampling Theory | Large Samples | Small Samples (t , F , χ^2)

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Contents

1	Sampling Theory	3
1.1	Terminologies	3
1.2	Errors in Sampling	4
1.3	Critical Region	4
1.4	Standard Normal Variate	4
1.5	One and Two Tailed Tests	5
2	Test of Significance of Large Samples	6
2.1	Procedure for Testing of Hypotheses	6
2.2	Critical Values for Other Levels of Significance	7
3	Test I : Difference between Sample Proportion p and Population Proportion P	7
3.1	Problem 1	7
3.2	Problem 2	8
3.3	Problem 3	9
4	Test II : Difference between Two Sample Proportions	10
4.1	Problem 1	10
4.2	Problem 2	11
5	Test III : Difference between Sample Mean $\bar{X}$ and Population Mean μ	12
5.1	Problem 1 – Mean height of students	13
5.2	Problem 2 – Breaking strength of cables	14
5.3	Problem 3 – 95% confidence limits	15
5.4	Problem 4 – Minimum sample size	17
6	Test IV : Difference between the Means of Two Samples	18
6.1	Note – The Three Cases	18
6.2	Problem 1 – Heights of English men and Americans	19

6.3	Problem 2 – Produce of wheat	20
6.4	Problem 3 – Marks of boys and girls	21
7	Test of Significance of Small Samples : the t-distribution	22
7.1	Properties of the t -distribution	22
8	t-Test I : Difference between Population Mean μ and Sample Mean $\bar{x}$	23
8.1	Problems	23
8.2	Problem 1 – Solution	23
8.3	Problem 2 – Solution	24
8.4	Exercises	25
9	t-Test II : Difference between Two Sample Means	26
9.1	Problem 1 – Length of life of electric bulbs	26
9.2	Problem 2 – Heights of Soldiers and Sailors	27
9.3	Problem 3 – Effect of coaching (paired data)	28
10	F-Test : Difference between Two Population Variances	30
10.1	Problem 1 – Two independent samples	31
10.2	Note – Samples from the same normal population	32
10.3	Problem 2 – Nicotine content of two samples	32
11	χ^2-Test (Chi-squared)	34
11.1	Test of Goodness of Fit	34
11.2	Problem 1 – Performance of four machines	35
11.3	Problem 2 – Examination results	36
11.4	Problem 3 – Boys and girls distribution	37
11.5	Problem 4 – Goodness of fit of a Poisson distribution	38
12	Exercises	40

1 Sampling Theory

What does this chapter deal with?

A salesman in a departmental store claims that at most 60 percent of the shoppers entering the store leave without making a purchase. . .

- How can we judge his claim, whether it is correct or not?
- We can determine it by collecting samples and its related result. Isn't it? Yes.
- This chapter deals with the study of inference of a collection using the information about its random member.

1.1 Terminologies

Basic terms

- **Population** – Collection of individuals.
- **Sample** – A finite subset of the population.
- **Statistic** – Statistical values computed from sample values (mean, variance etc.).

Testing of Hypothesis

Testing of known information of the population with the random sample drawn from the population, whether it can be accepted or rejected.

Hypotheses

- **Null Hypothesis** (H_0) – Assumes that there is no significance between sample statistic and population parameter.
- **Alternative Hypothesis** (H_1) – Assumes that there is some significance between sample statistic and population parameter.

Tests

- **Test of Hypothesis** – A procedure of deciding whether to accept or reject H_0 .
- **Test of Significance** – A procedure of testing whether the difference between the population parameter θ_0 and the sample statistic θ is significant or not.

1.2 Errors in Sampling

Standard Error

S.D. of the sampling distribution.

Type I Error

Rejecting H_0 when it is really true. (Producer's Risk)

Type II Error

Accepting H_0 when it is really false. (Consumer's Risk)

1.3 Critical Region

Critical region (Region of rejection)

A region corresponding to the sample statistic θ which leads to the rejection of H_0 .

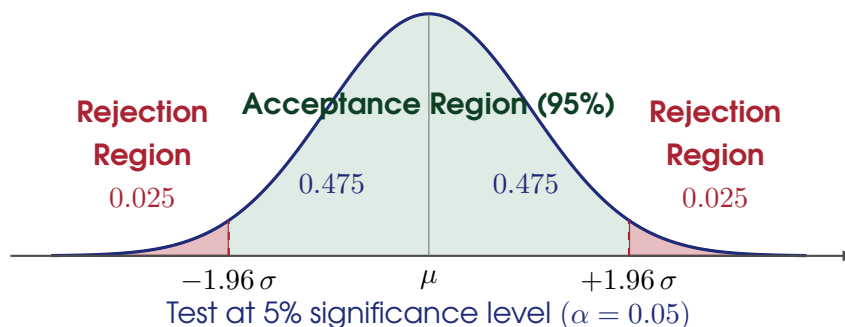
Region of acceptance

The region complementary to the critical region.

Note

In a sampling distribution many statistics tend to a normal distribution.

Critical region – Two tailed test at 5% LOS



1.4 Standard Normal Variate

Standard normal variate

If t is a test statistic $\sim N(E(t), SE(t))$ then $Z = \frac{t - E(t)}{SE(t)}$ is a standard normal variate, and

$$P(-1.96 < Z < 1.96) = P(|Z| < 1.96) = 95\%.$$

Decision at 5% LOS

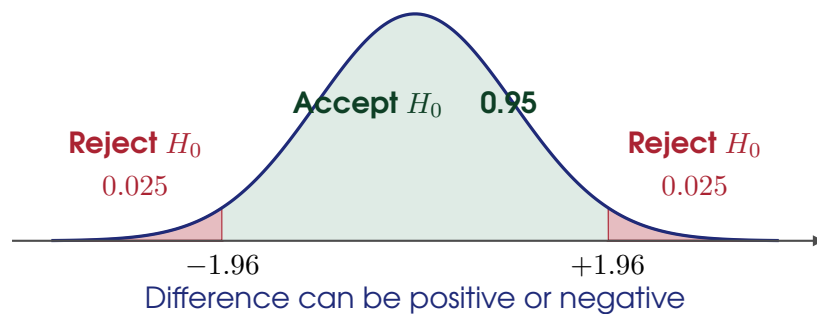
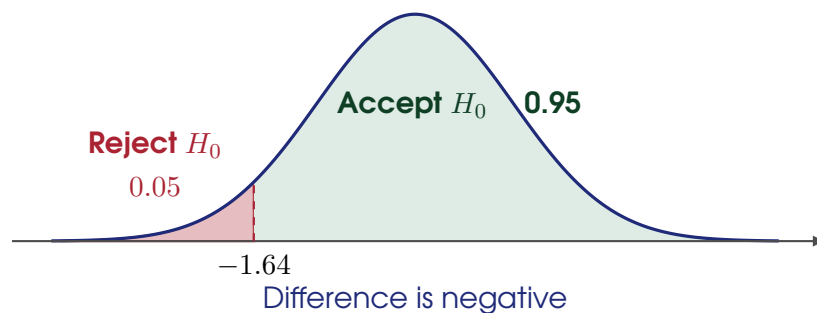
- Hence if $\left| \frac{t - E(t)}{SE(t)} \right| < 1.96$, we can say that there is no significance between t and $E(t)$.
- i.e., we can accept H_0 with 95% confidence limit or 5% level of significance (LOS).
- If $|Z| > 1.96$, we can reject H_0 at 5% LOS.

1.5 One and Two Tailed Tests**One and two tailed tests**

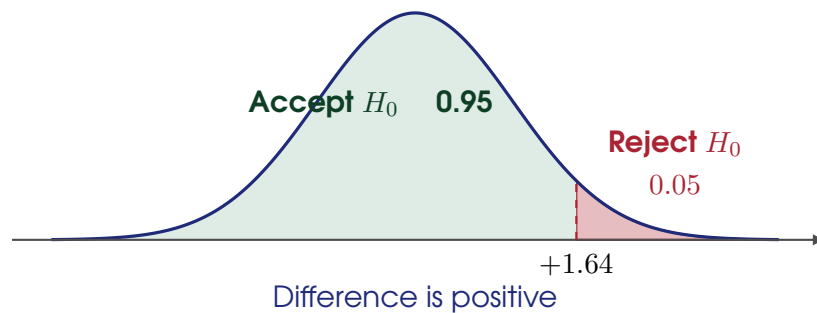
For the population parameter θ_0 and the sample statistic θ we set up the null hypothesis as $H_0 : \theta = \theta_0$; then the alternative hypothesis is any one of the following (depending on the nature of the problem).

The three possibilities

- (i) $H_1 : \theta \neq \theta_0 \Rightarrow$ **two tailed test**
- (ii) $H_1 : \theta < \theta_0 \Rightarrow$ **left tailed test**
- (iii) $H_1 : \theta > \theta_0 \Rightarrow$ **right tailed test**

Two tailed test**Left tailed test**

Right tailed test



2 Test of Significance of Large Samples

Large samples

- Large samples – Sample size exceeds 30 ($n > 30$).
- The sampling distribution approximately tends to a normal distribution.

Critical value

- Critical value – A value which divides the critical and the acceptance region.
- Depending on the tailed test and the LOS (α), the critical value varies.

Critical values for LOS $\alpha = 5\%$

Test	Two-tailed	Left-tailed	Right-tailed
Critical value (z_α)	1.96	-1.64	1.64

2.1 Procedure for Testing of Hypotheses

Step by step

1. Identify the appropriate test based on the given values.
2. Set up H_0 and H_1 .
3. Identify the tailed test based on H_1 .
4. Calculate the test statistic z .
5. Identify the critical value z_α .
6. If $|z| < |z_\alpha|$, accept H_0 ; otherwise reject H_0 (accept H_1).
7. Write the conclusion in words.

2.2 Critical Values for Other Levels of Significance

Critical values for large samples (from the normal table)

Nature of test	z_α (5%)	z_α (2%)	z_α (1%)
Two-tailed	1.96	2.33 (or 2.32)	2.57 (or 2.58)
Right-tailed	1.64 (or 1.645)	2.05	2.33
Left-tailed	-1.64 (or -1.645)	-2.05	-2.33

Reading the normal table

For $\alpha = 2\%$ (right tailed) the area between $z = 0$ and $z = z_\alpha$ is 0.48 (48%), leaving 2% in the tail $\Rightarrow z_\alpha = 2.05$. For $\alpha = 1\%$ the corresponding area is 0.495 (49.5%) $\Rightarrow z_\alpha = 2.33$.

3 Test I : Difference between Sample Proportion p and Population Proportion P

Test I : Test statistic

The test statistic for the test of significance of the difference between the sample proportion (p) and the population proportion (P) is

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}, \quad Q = 1 - P.$$

3.1 Problem 1

Problem 1

A salesman in a departmental store claims that at most 60 percent of the shoppers entering the store leave without making a purchase. A random sample of 50 shoppers showed that 35 of them left without making a purchase. Are these sample results consistent with the claim of the salesman?

Solution – Given

$$P = 60\% = 0.6 \Rightarrow Q = 1 - P = 0.4, \quad n = 50, \quad p = \frac{35}{50} = \frac{7}{10} = 0.7$$

Hypotheses**Null Hyp** (H_0) : $p = P$ (No significant difference between the sample proportion p and the population proportion $P \Rightarrow$ sample results are consistent with the claim of the salesman.)**Alt. Hyp** (H_1) : $p \neq P$ (There is some significant difference between p and $P \Rightarrow$ sample results are inconsistent with the claim of the salesman.) $\Rightarrow$ **The test is a two tailed test.****Test statistic**

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.7 - 0.6}{\sqrt{\frac{0.6(0.4)}{50}}} \Rightarrow \boxed{z = 1.45}$$

Critical valueFor a two tailed test and for LOS $\alpha = 5\%$,

$\boxed{z_\alpha = 1.96}$

Comparison

$|z| = 1.45 < 1.96 = |z_\alpha| \Rightarrow |z| < |z_\alpha| \Rightarrow \text{Accept } H_0$

ConclusionSample results are **consistent** with the claim of the salesman.**3.2 Problem 2****Problem 2**

The fatality rate of typhoid patients is believed to be 17.26%. In a certain year 640 patients suffering from typhoid were treated in a metropolitan hospital and only 63 patients died. Can you consider the hospital efficient?

Solution – Given

$$P = 17.26\% = 0.1726 \Rightarrow Q = 1 - P \Rightarrow \boxed{Q = 0.8274}, \quad n = 640,$$

$$p = \frac{63}{640} = 0.0984$$

Hypotheses**Null Hyp** (H_0) : $p = P$ (No significant difference between p and $P \Rightarrow$ the hospital is inefficient.)

Alt. Hyp (H_1) : $p < P$

(There is some significance between p and $P \Rightarrow$ the hospital is efficient.)

$\Rightarrow$ **The test is a left tailed test.**

Test statistic

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.0984 - 0.1726}{\sqrt{\frac{(0.1726)(0.8274)}{640}}} \Rightarrow \boxed{z = -4.967}$$

Critical value

For a left tailed test and for LOS $\alpha = 5\%$,

$$\boxed{z_\alpha = -1.64}$$

Comparison

$$|z| = 4.967 > 1.64 = |z_\alpha| \Rightarrow |z| > |z_\alpha| \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion

The hospital is efficient.

3.3 Problem 3

Problem 3

A manufacturer of fluorescent bulbs claims that on an average 2% of his products are defective. A random sample of 400 items contains 13 defectives. On the basis of the sample can you support the manufacturer's claim at 5% LOS?

Solution – Given

$$P = 0.02 \Rightarrow Q = 1 - P = 0.98, \quad p = \frac{13}{400} = 0.0325, \quad n = 400$$

Hypotheses

Null Hyp (H_0) : $p = P$

(No significant difference between p and $P \Rightarrow$ we can support the manufacturer's claim.)

Alt. Hyp (H_1) : $p \neq P$

(There is some significant difference between p and $P \Rightarrow$ we can't support the manufacturer's claim.)

$\Rightarrow$ **The test is a two tailed test.**

Test statistic

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.0325 - 0.02}{\sqrt{\frac{(0.02)(0.98)}{400}}} \Rightarrow \boxed{z = 1.7857}$$

Critical value

For a two tailed test and for LOS $\alpha = 5\%$, $\boxed{z_{\alpha} = 1.96}$

Comparison

$$|z| = 1.7857 < 1.96 = |z_{\alpha}| \Rightarrow |z| < |z_{\alpha}| \Rightarrow \text{Accept } H_0$$

Conclusion

We can support the manufacturer's claim.

4 Test II : Difference between Two Sample Proportions

Test II : Test statistic

$$z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}, \quad P \approx \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \quad Q = 1 - P.$$

4.1 Problem 1

Problem 1

In a city A, 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the two proportions significant?

Solution – Given

$$n_1 = 900, \quad p_1 = 20\% = 0.2 \quad n_2 = 1600, \quad p_2 = 18.5\% = 0.185$$

Pooled proportion

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{900(0.2) + 1600(0.185)}{900 + 1600}$$

$$\Rightarrow \boxed{P = 0.1904} \quad \Rightarrow \boxed{Q = 1 - P = 0.8096}$$

Hypotheses**Null Hyp** (H_0) : $p_1 = p_2$ (No significant difference between the two sample proportions p_1 and p_2 .)**Alt. Hyp** (H_1) : $p_1 \neq p_2$

(There is some significant difference between the two sample proportions.)

 $\Rightarrow$ **The test is a two tailed test.****Test statistic**

$$z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.2 - 0.185}{\sqrt{(0.1904)(0.8096) \left(\frac{1}{900} + \frac{1}{1600} \right)}} \Rightarrow \boxed{z = 0.9169}$$

Critical valueFor a two tailed test and for LOS $\alpha = 5\%$,

$\boxed{z_\alpha = 1.96}$

Comparison

$$|z| = 0.9169 < 1.96 = |z_\alpha| \Rightarrow |z| < |z_\alpha| \Rightarrow \text{Accept } H_0$$

Conclusion

There is no significant difference between the two sample proportions p_1 and p_2 .

4.2 Problem 2**Problem 2**

Before an increase in excise duty on tea, 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increase in the duty, 800 persons were tea drinkers in a sample of 1200 persons. Test whether there is a significant decrease in the consumption of tea after the increase in excise duty at 5% LOS.

Solution – Given

$$p_1 = \frac{800}{1000} \Rightarrow \boxed{p_1 = 0.8}, \quad p_2 = \frac{800}{1200} \Rightarrow \boxed{p_2 = 0.666}, \quad n_1 = 1000, \quad n_2 = 1200$$

Pooled proportion

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{1000(0.8) + 1200(0.666)}{2200}$$

$$\Rightarrow \boxed{P = 0.7272} \quad \Rightarrow \quad \boxed{Q = 0.2727}$$

Hypotheses

Null Hyp (H_0) : $p_1 = p_2$

(No significant difference between the two sample proportions $\Rightarrow$ there is no significant decrease in the consumption of tea.)

Alt. Hyp (H_1) : $p_1 > p_2$

(There is some significant difference between the two sample proportions $\Rightarrow$ there is some significant decrease in the consumption of tea after the increase in excise duty.)

$\Rightarrow$ **The test is a right tailed test.**

Test statistic

$$z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad \Rightarrow \quad \boxed{z = 6.975}$$

Critical value

For a right tailed test and for LOS $\alpha = 5\%$,

$$\boxed{z_\alpha = 1.64}$$

Comparison

$$|z| = 6.975 > 1.64 = |z_\alpha| \Rightarrow |z| > |z_\alpha| \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion

There is some significant decrease in the consumption of tea after the increase in excise duty.

5 Test III : Difference between Sample Mean $\bar{X}$ and Population Mean μ

Test III : Test statistic

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Note

If σ is not known, the sample S.D. s can be used in its place, as s is nearly equal to σ when n is large.

Working procedure

1. Write down the **given** data: $\bar{x}$, μ , σ (or s) and n .
2. State the **Null Hypothesis** H_0 (with its meaning in words).
3. State the **Alternative Hypothesis** H_1 (with its meaning in words).
4. Decide whether the test is **two-tailed**, **right-tailed** or **left-tailed**.
5. Compute the **test statistic** $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$.
6. Read the **critical value** z_α for the given level of significance (LOS).
7. Compare $|z|$ with $|z_\alpha|$ and **accept** or **reject** H_0 .
8. Write the **conclusion** in words.

Critical values z_α for large samples

Nature of test	$\alpha = 1\%$	$\alpha = 2\%$	$\alpha = 5\%$
Two-tailed	2.58	2.33	1.96
Right-tailed	2.33	2.05	1.645
Left-tailed	-2.33	-2.05	-1.645

(Values read from the normal probability table.)

5.1 Problem 1 – Mean height of students**Problem 1**

A sample of 100 students is taken from a large population. The mean height of students in this sample is 160 cm. Can it be reasonably regarded that, in the population, the mean height is 165 cm and S.D. is 10 cm?

Solution – Given

$\bar{x}$ = Mean height in the sample = 160

n = 100

μ = Mean height in the population = 165

σ = S.D. in the population = 10

Hypotheses**Null Hyp. (H_0):** $\bar{x} = \mu$ (No significant difference between the sample mean $\bar{x}$ and the population mean $\mu \Rightarrow$ the sample can be regarded as drawn from the given population.)**Alt. Hyp. (H_1):** $\bar{x} \neq \mu$ (There is some significant difference between $\bar{x}$ and $\mu \Rightarrow$ the sample cannot be regarded as drawn from the given population.) $\Rightarrow$ **The test is a two-tailed test.****Test statistic**

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{160 - 165}{\frac{10}{\sqrt{100}}} = \frac{-5}{1} \Rightarrow \boxed{z = -5}$$

Critical valueFor a two-tailed test and for LOS $\alpha = 5\%$,

$$\boxed{z_\alpha = 1.96}$$

Comparison

$$|z| = 5 > 1.96 = |z_\alpha| \Rightarrow |z| > |z_\alpha| \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

ConclusionThe difference between $\bar{x}$ and μ is significant, i.e. $\bar{x} \neq \mu$. Hence it is *not* statistically correct to assume that $\mu = 165$.**5.2 Problem 2 – Breaking strength of cables****Problem 2**

The mean breaking strength of the cables supplied by a manufacturer is 1800 with a S.D. of 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cable has increased. In order to verify this claim, a sample of 50 cables is tested and it is found that the mean breaking strength is 1850. Can we support the claim at 1% level of significance?

Solution – Given

$\bar{x}$ = Mean strength of the cable in the sample = 1850

n = 50

μ = Mean strength in the population = 1800

σ = 100, LOS α = 1%

Hypotheses

Null Hyp. (H_0): $\bar{x} = \mu$

(No significant difference between $\bar{x}$ and $\mu \Rightarrow$ there is *no* increase in the breaking strength; the claim cannot be supported.)

Alt. Hyp. (H_1): $\bar{x} > \mu$

(There is some significant difference between $\bar{x}$ and $\mu \Rightarrow$ the breaking strength has increased; the claim can be supported.)

$\Rightarrow$ **The test is a one-tailed (right-tailed) test.**

Test statistic

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{1850 - 1800}{\frac{100}{\sqrt{50}}} \Rightarrow \boxed{z = 3.54}$$

Critical value

For a right-tailed test and for LOS $\alpha = 1\%$,

$$\boxed{z_\alpha = 2.33}$$

Comparison

$$|z| = 3.54 > 2.33 = |z_\alpha| \Rightarrow |z| > |z_\alpha| \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion

The difference between $\bar{x}$ and μ is significant at the 1% level. Based on the sample data, we may **support the claim** of an increase in breaking strength.

5.3 Problem 3 – 95% confidence limits**Problem 3**

The mean value of a random sample of 60 items was found to be 145 with a S.D. of 40. Find the 95% confidence limits for the population mean.

Solution – Given and formula**Given** $\bar{x} = 145, \quad s = 40, \quad n = 60$ 95% confidence limits for μ are given by

$$\frac{|\bar{x} - \mu|}{\frac{\sigma}{\sqrt{n}}} \leq 1.96$$

Since the population S.D. σ too is not given, we approximate it by the sample S.D. s :

$$\frac{|\bar{x} - \mu|}{\frac{s}{\sqrt{n}}} \leq 1.96$$

Hypotheses**Null Hyp. (H_0):** $\bar{x} = \mu$

(No significant difference between the sample mean $\bar{x}$ and the population mean $\mu \Rightarrow$ the assumed value of μ is acceptable and lies within the confidence limits.)

Alt. Hyp. (H_1): $\bar{x} \neq \mu$

(There is some significant difference between $\bar{x}$ and $\mu \Rightarrow$ the assumed value of μ is not acceptable and lies outside the confidence limits.)

$\Rightarrow$ **The test is a two-tailed test**, and for 95% confidence (LOS $\alpha = 5\%$) the critical value is $z_\alpha = 1.96$

Confidence limits

$$\begin{aligned} \bar{x} - 1.96 \frac{s}{\sqrt{n}} &\leq \mu \leq \bar{x} + 1.96 \frac{s}{\sqrt{n}} \\ 145 - \frac{1.96 \times 40}{\sqrt{60}} &\leq \mu \leq 145 + \frac{1.96 \times 40}{\sqrt{60}} \\ 134.9 &\leq \mu \leq 155.1 \end{aligned}$$

Conclusion

The 95% confidence limits for the population mean are 134.9 and 155.1.

5.4 Problem 4 – Minimum sample size

Problem 4

The guaranteed average life of a certain type of electric light bulbs is 1000 hours with a standard deviation of 125 hours. It is decided to sample the output so as to ensure that 90 percent of the bulbs do not fall short of the guaranteed average by more than 2.5 percent. What must be the minimum size of the sample?

Solution – Given

Given $\mu = 1000$ hours, $\sigma = 125$ hours.

The bulbs must not fall short of the guaranteed average by more than 2.5%:

$$\begin{aligned}\bar{x} &> 1000 - 2.5\% \text{ of } 1000 \\ \Rightarrow \bar{x} &> 1000 - 25 = 975\end{aligned}$$

Hypotheses

Null Hyp. (H_0): $\bar{x} = \mu$

(No significant difference between $\bar{x}$ and $\mu \Rightarrow$ the bulbs do *not* fall short of the guaranteed average life.)

Alt. Hyp. (H_1): $\bar{x} < \mu$

(There is some significant difference between $\bar{x}$ and $\mu \Rightarrow$ the bulbs fall short of the guaranteed average life by more than the permitted 2.5%.)

$\Rightarrow$ **The test is a one-tailed (left-tailed) test.**

Setting up the probability

Now

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{975 - 1000}{\frac{125}{\sqrt{n}}} = -\frac{\sqrt{n}}{5}$$

It is given that 90 percent of the bulbs satisfy this, i.e.

$$P\left(z > -\frac{\sqrt{n}}{5}\right) = 0.90$$

Minimum sample size

$$\Rightarrow P\left(0 < z < \frac{\sqrt{n}}{5}\right) = 0.40$$

$$\Rightarrow \frac{\sqrt{n}}{5} = 1.28$$

$$\Rightarrow n = 25 \times (1.28)^2 = 41 \text{ (approximately)}$$

Conclusion

The minimum size of the sample must be 41.

6 Test IV : Difference between the Means of Two Samples

Test IV : Set-up

Let $\bar{X}_1$ and $\bar{X}_2$ be the means of two large independent samples of sizes n_1 and n_2 drawn from two populations (normal or non-normal) with the same mean μ and variances σ_1^2 and σ_2^2 respectively.

Test IV : Test statistic

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

6.1 Note – The Three Cases

Case 1 : samples from the same population, $\sigma_1 = \sigma_2 = \sigma$ known

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Case 2 : $\sigma_1 \neq \sigma_2$ and both unknown

The sample standard deviations s_1 and s_2 are used in their place:

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Case 3 : $\sigma_1 = \sigma_2 = \sigma$, but σ not known

σ is approximated by the pooled standard deviation

$$\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}, \quad \text{and then} \quad z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

6.2 Problem 1 – Heights of English men and Americans**Problem 1**

A simple sample of heights of 6400 English men has a mean of 170 cm and S.D. of 6.4 cm, while a simple sample of heights of 1600 Americans has a mean of 172 cm and a S.D. of 6.3 cm. Do the data indicate that Americans are, on the average, taller than the English men? Test at 1% level of significance.

Solution – Given

$$\bar{X}_1 = 170, \quad n_1 = 6400, \quad s_1 = 6.4 \qquad \bar{X}_2 = 172, \quad n_2 = 1600, \quad s_2 = 6.3$$

Hypotheses

Null Hyp. (H_0): $\bar{X}_1 = \bar{X}_2$

(No significant difference between the two sample means $\Rightarrow$ the Americans are *not* taller than the English men.)

Alt. Hyp. (H_1): $\bar{X}_1 < \bar{X}_2$

(There is some significant difference between $\bar{X}_1$ and $\bar{X}_2 \Rightarrow$ the Americans are, on the average, taller than the English men.)

$\Rightarrow$ **The test is a left tailed test.**

Test statistic

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{170 - 172}{\sqrt{\frac{(6.4)^2}{6400} + \frac{(6.3)^2}{1600}}}$$

$$\Rightarrow \boxed{z = -11.32}$$

Critical value

For a left tailed test and for LOS $\alpha = 1\%$,

$$\boxed{z_\alpha = -2.33}$$

Comparison

$$|z| = 11.32 > 2.33 = |z_\alpha| \Rightarrow |z| > |z_\alpha| \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion

The Americans are, on the average, taller than the English men.

6.3 Problem 2 – Produce of wheat**Problem 2**

The mean produce of wheat from a sample of 100 fields comes to 200 kg per acre, and another sample of 150 fields gives a mean of 220 kg per acre. Assuming the S.D. of the yield is 11 kg for the universe, test if there is a significant difference between the means of the samples.

Solution – Given

$$n_1 = 100, \quad \bar{X}_1 = 200 \qquad n_2 = 150, \quad \bar{X}_2 = 220 \qquad \sigma = 11$$

Hypotheses

Null Hyp. (H_0): $\bar{X}_1 = \bar{X}_2$

(No significant difference between the means of the two samples.)

Alt. Hyp. (H_1): $\bar{X}_1 \neq \bar{X}_2$

(There is some significant difference between the means of the two samples.)

$\Rightarrow$ The test is a two tailed test.

Test statistic (σ is known $\Rightarrow$ Case 1)

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{200 - 220}{11 \sqrt{\frac{1}{100} + \frac{1}{150}}}$$

$$\Rightarrow \boxed{z = -14.08}$$

Critical value

For a two tailed test and for LOS $\alpha = 5\%$,

$$\boxed{z_\alpha = 1.96}$$

Comparison

$$|z| = 14.08 > 1.96 = |z_\alpha| \Rightarrow |z| > |z_\alpha| \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion

There is a significant difference between the means of the two samples.

6.4 Problem 3 – Marks of boys and girls

Problem 3

The average marks scored by 32 boys is 72 with a S.D. of 8, while that for 36 girls is 70 with a S.D. of 6. Test at 1% level of significance whether the boys perform better than the girls.

Solution – Given

$$\bar{X}_1 = 72, \quad n_1 = 32, \quad s_1 = 8 \qquad \bar{X}_2 = 70, \quad n_2 = 36, \quad s_2 = 6$$

Hypotheses

Null Hyp. (H_0): $\bar{X}_1 = \bar{X}_2$

(No significant difference between the two sample means $\Rightarrow$ the boys do *not* perform better than the girls.)

Alt. Hyp. (H_1): $\bar{X}_1 > \bar{X}_2$

(There is some significant difference between $\bar{X}_1$ and $\bar{X}_2 \Rightarrow$ the boys perform better than the girls.)

$\Rightarrow$ **The test is a right tailed test.**

Test statistic

$$z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{72 - 70}{\sqrt{\frac{8^2}{32} + \frac{6^2}{36}}} = \frac{2}{\sqrt{2.983}} = \frac{2}{1.726}$$

$$\Rightarrow \boxed{z = 1.16}$$

Critical value

For a right tailed test and for LOS $\alpha = 1\%$,

$$\boxed{z_\alpha = 2.33}$$

Comparison

$$|z| = 1.16 < 2.33 = |z_\alpha| \Rightarrow |z| < |z_\alpha| \Rightarrow \text{Accept } H_0$$

Conclusion

We cannot conclude that the boys perform better than the girls.

7 Test of Significance of Small Samples : the t -distribution

Small samples

- Small samples – sample size less than 30 ($n < 30$).
- The sampling distribution approximately follows any one of the following: (i) t -distribution, (ii) F -distribution, (iii) χ^2 -distribution.

t -distribution : probability density function

$$f(t) = \frac{1}{\sqrt{v} \beta \left(\frac{v}{2}, \frac{1}{2} \right)} \left(1 + \frac{t^2}{v} \right)^{-\left(\frac{v+1}{2} \right)}, \quad -\infty < t < \infty$$

Degrees of freedom v

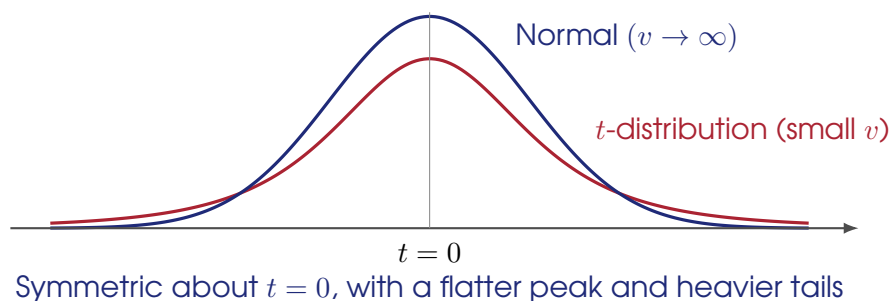
The difference between the number of independent variables and the number of restrictions, when a statistic is calculated.

7.1 Properties of the t -distribution

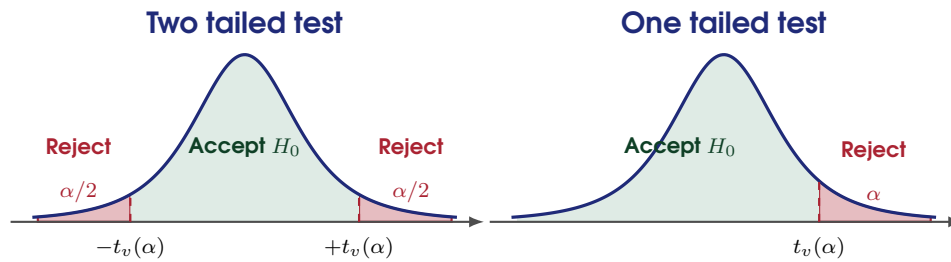
Properties

- Mean = 0 and variance = $\frac{v}{v-2}$.
- It is a bell shaped curve, symmetric about $t = 0$.
- It tends to the normal distribution for large values of v .
- Critical values are denoted by $t_v(\alpha)$ – they depend on both the degrees of freedom v and the level of significance α .

The t -distribution compared with the normal distribution



Critical regions of the t -distribution



8 t -Test I : Difference between Population Mean μ and Sample Mean $\bar{x}$

t -Test I : Test statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$$

where s is the S.D. of the sample and $v = n - 1$ is the number of degrees of freedom.

8.1 Problems

Problem 1

The mean weekly sales of soap bars in a departmental store was found to be 145 bars per store. After an advertisement campaign, the mean weekly sales of soap bars in 22 departmental stores was found to be 155 bars per store with a S.D. of 7. Was the advertisement campaign successful?

Problem 2

Is the sample values 45, 51, 55, 58, 59, 46, 48, 60, 62, 57 differs significantly from the population mean of 56?

8.2 Problem 1 – Solution

Given

$$\mu = 145, \quad \bar{x} = 155, \quad s = 7, \quad n = 22$$

Hypotheses**Null Hyp. (H_0):** $\bar{x} = \mu$ (No significant difference $\Rightarrow$ the advertisement campaign was not successful.)**Alt. Hyp. (H_1):** $\bar{x} > \mu$ (There is significance $\Rightarrow$ the advertisement campaign was successful.) $\Rightarrow$ **The test is a one tailed test.****Test statistic**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{155 - 145}{\frac{7}{\sqrt{22-1}}}$$

$$\Rightarrow \boxed{t = 6.581}$$

Critical valueOne tailed test, $\alpha = 5\%$ LOS and $v = n - 1 = 21$ degrees of freedom $\Rightarrow$

$$\boxed{t_{21}(0.05) = 1.721} \quad (\text{from the } t\text{-table})$$

Comparison

$$|t| = 6.581 > 1.721 = t_v(\alpha) \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion**The advertisement campaign was successful.****8.3 Problem 2 – Solution****Given** $\mu = 56, \quad n = 10, \quad \text{sample values : } 45, 51, 55, 58, 59, 46, 48, 60, 62, 57$ **Sample mean**

$$\bar{x} = \frac{\sum x_i}{n} = \frac{45 + 51 + 55 + 58 + 59 + 46 + 48 + 60 + 62 + 57}{10} = \frac{541}{10}$$

$$\Rightarrow \boxed{\bar{x} = 54.1}$$

Variance and standard deviation

$$s^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2 = \frac{29609}{10} - (54.1)^2 = 2960.9 - 2926.81 \Rightarrow \boxed{s^2 = 34.09}$$

$$s = \sqrt{34.09} \Rightarrow \boxed{s = 5.84}$$

Hypotheses**Null Hyp. (H_0):** $\bar{x} = \mu$

(No significant difference between the sample mean and the population mean.)

Alt. Hyp. (H_1): $\bar{x} \neq \mu$

(There is some significant difference between the sample mean and the population mean.)

 $\Rightarrow$ **The test is a two tailed test.****Test statistic ($v = n - 1 = 9$ degrees of freedom)**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{54.1 - 56}{\frac{5.84}{\sqrt{9}}} \Rightarrow t = -0.976$$

Critical valueTwo tailed test, $\alpha = 5\%$ LOS and $v = 9$ degrees of freedom $\Rightarrow t_9(0.05) = 2.262$ **Comparison**

$$|t| = 0.976 < 2.262 = |t_v(\alpha)| \Rightarrow |t| < |t_v(\alpha)| \Rightarrow \text{Accept } H_0$$

Conclusion**There is no significant difference between the sample mean and the population mean.****8.4 Exercises****Exercise 1**

A random sample of 16 values from a normal population showed a mean of 41.5 inches and sum of squares of deviations from this mean equal to 135 square inches. Show that the assumption of a mean of 43.5 for the population is not reasonable.

Exercise 2

Suppose that the annual rainfall x at a certain place is normally distributed with $E(x) = 30$. Data for the past eight years are given below. Test $H_0 : \mu = 30.0$ against $H_1 : \mu \neq 30.0$ at 5% level.

Year	1983	1984	1985	1986	1987	1988	1989	1990
x	34.1	33.7	27.4	31.1	30.9	35.2	28.4	32.1

9 *t*-Test II : Difference between Two Sample Means

Set-up

Let $\bar{X}_1$ and $\bar{X}_2$ be two sample means of sizes n_1 and n_2 respectively. The test statistic takes one of three forms, depending on the sample sizes and on whether the samples are independent or paired.

Case (i) : $n_1 \neq n_2$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \quad v = n_1 + n_2 - 2$$

where s_1^2 and s_2^2 are the variances of the two samples.

Case (ii) : $n_1 = n_2 = n$, independent samples

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2 + s_2^2}{n - 1}}}, \quad v = 2n - 2$$

Case (iii) : $n_1 = n_2 = n$, dependent (paired) samples

$$t = \frac{\bar{d}}{\frac{s}{\sqrt{n - 1}}}, \quad v = n - 1$$

where $d_i = x_i - y_i$, $\bar{d} = \frac{\sum d_i}{n}$ and $s^2 = \frac{\sum d_i^2}{n} - (\bar{d})^2$.

9.1 Problem 1 – Length of life of electric bulbs

Problem 1

The sample values of two types of electric bulbs were tested for the length of life and the following data were obtained. Is the difference in the means sufficient to warrant that Type II bulbs are superior to Type I bulbs?

Sample	Size	Mean (hrs)	S.D. (hrs)
I	8	1234	36
II	7	1236	40

Solution – Given ($n_1 \neq n_2 \Rightarrow$ **Case (i)**)

$$n_1 = 8, \quad \bar{X}_1 = 1234, \quad s_1 = 36$$

$$n_2 = 7, \quad \bar{X}_2 = 1236, \quad s_2 = 40$$

Hypotheses

Null Hyp. (H_0): $\bar{X}_1 = \bar{X}_2$

(No significant difference between the two sample means $\Rightarrow$ Type II bulbs are *not* superior.)

Alt. Hyp. (H_1): $\bar{X}_1 < \bar{X}_2$

(There is significant difference $\Rightarrow$ Type II bulbs are superior to Type I bulbs.)

$\Rightarrow$ **The test is a one tailed test.**

Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{1234 - 1236}{\sqrt{\left(\frac{8(36)^2 + 7(40)^2}{13}\right) \left(\frac{1}{8} + \frac{1}{7}\right)}} = \frac{-2}{21.08}$$

$$\Rightarrow \boxed{t = -0.095}$$

Critical value

One tailed test, $\alpha = 5\%$ and $v = n_1 + n_2 - 2 = 8 + 7 - 2 = 13$ degrees of freedom

$$\Rightarrow \boxed{t_{13}(0.05) = 1.771}$$

Comparison

$$|t| = 0.095 < 1.771 = t_v(\alpha) \Rightarrow |t| < t_v(\alpha) \Rightarrow \textbf{Accept } H_0$$

Conclusion

Type II bulbs are not superior to Type I bulbs.

9.2 Problem 2 – Heights of Soldiers and Sailors**Problem 2**

The mean height and S.D. of 8 randomly chosen Soldiers are 166.9 cm and 8.29 cm respectively. The corresponding values of 8 randomly chosen Sailors are 170.3 cm and 8.50 cm respectively. Based on this data, can we conclude that, in general, Soldiers are shorter than Sailors?

Solution – Given ($n_1 = n_2 = 8$, independent $\Rightarrow$ **Case (ii)**)

$$\bar{X}_1 = 166.9, \quad s_1 = 8.29$$

$$\bar{X}_2 = 170.3, \quad s_2 = 8.50$$

Hypotheses**Null Hyp. (H_0):** $\bar{X}_1 = \bar{X}_2$ (No significant difference between the two sample means $\Rightarrow$ Soldiers are *not* shorter than Sailors.)**Alt. Hyp. (H_1):** $\bar{X}_1 < \bar{X}_2$ (There is some significance $\Rightarrow$ Soldiers are shorter than Sailors.) $\Rightarrow$ **The test is a one tailed test.****Test statistic**

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}} = \frac{166.9 - 170.3}{\sqrt{\frac{(8.29)^2 + (8.50)^2}{7}}} \Rightarrow \boxed{t = -0.758}$$

Critical valueOne tailed test, $\alpha = 5\%$ and $v = 2n - 2 = 14$ degrees of freedom $\Rightarrow$

$$\boxed{t_{14}(0.05) = 1.761}$$

Comparison

$$|t| = 0.758 < 1.761 = |t_v(\alpha)| \Rightarrow |t| < |t_v(\alpha)| \Rightarrow \text{Accept } H_0$$

Conclusion**Soldiers are not shorter than Sailors in general.****9.3 Problem 3 – Effect of coaching (paired data)****Problem 3**

The following data relate to the marks obtained by 11 students in one test at the beginning of the year and another at the end of the year after intensive coaching. Do the data indicate that the students have been benefited by coaching?

Test I (x_i)	19	23	16	24	17	18	20	18	21	19	20
Test II (y_i)	17	24	20	24	20	22	20	20	18	22	19

Solution – Given ($n_1 = n_2 = n = 11$, paired $\Rightarrow$ Case (iii))

The data values are dependent, since they have been taken from the same set of students. Let $d_i = x_i - y_i$:

d_i	2	-1	-4	0	-3	-4	0	-2	3	-3	1
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Mean, variance and standard deviation of the differences

$$\bar{d} = \frac{\sum d_i}{n} = \frac{-11}{11} \Rightarrow \boxed{\bar{d} = -1}$$

$$\sum d_i^2 = 2^2 + (-1)^2 + (-4)^2 + 0^2 + (-3)^2 + (-4)^2 + 0^2 + (-2)^2 + 3^2 + (-3)^2 + 1^2 = 69$$

$$s^2 = \frac{\sum d_i^2}{n} - (\bar{d})^2 = \frac{69}{11} - 1 \Rightarrow \boxed{s^2 = 5.27} \Rightarrow \boxed{s = 2.296}$$

Hypotheses

Null Hyp. (H_0): $\bar{d} = 0$

(No significant difference between the two sample means $\Rightarrow$ the students have *not* benefited by coaching.)

Alt. Hyp. (H_1): $\bar{d} < 0$

(There is a significant decrease in the differences $\Rightarrow$ the students have benefited by coaching.)

$\Rightarrow$ **The test is a one tailed test.**

Paired t -test statistic

$$t = \frac{\frac{\bar{d}}{s}}{\frac{1}{\sqrt{n-1}}} = \frac{-1}{\frac{2.296}{\sqrt{10}}} \Rightarrow \boxed{t = -1.377}$$

Critical value

One tailed test, $\alpha = 5\%$ and $v = n - 1 = 10$ degrees of freedom $\Rightarrow$

$$\boxed{t_{10}(0.05) = 1.812}$$

Comparison

$$|t| = 1.377 < 1.812 = t_v(\alpha) \Rightarrow |t| < t_v(\alpha) \Rightarrow \textbf{Accept } H_0$$

Conclusion

The students have not been benefited by coaching.

10 F-Test : Difference between Two Population Variances

Uses of the F-distribution

- To test the significant difference between two population variances given by two samples.
- To test whether two samples are taken from the same normal population.

Estimates of the population variances

Let s_1^2 and s_2^2 be the variances of two samples of sizes n_1 and n_2 respectively.

Then the population variances are estimated by

$$\hat{\sigma}_1^2 = \left(\frac{n_1}{n_1 - 1} \right) s_1^2, \quad \hat{\sigma}_2^2 = \left(\frac{n_2}{n_2 - 1} \right) s_2^2$$

Hypotheses

Null Hyp. (H_0): $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$

(No significant difference between the population variances.)

Alt. Hyp. (H_1): $\hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$

(There is some significant difference between the population variances.)

Test statistic – always the larger estimate on top

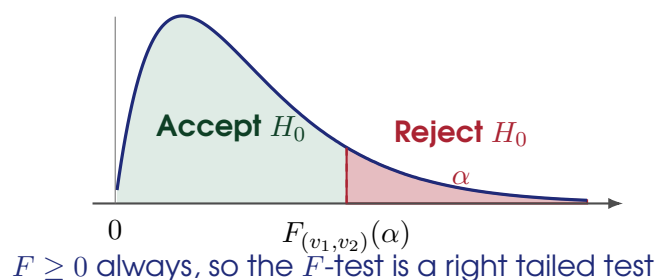
$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} \text{ if } \hat{\sigma}_1^2 > \hat{\sigma}_2^2, \quad (v_1, v_2) = (n_1 - 1, n_2 - 1)$$

$$F = \frac{\hat{\sigma}_2^2}{\hat{\sigma}_1^2} \text{ if } \hat{\sigma}_2^2 > \hat{\sigma}_1^2, \quad (v_1, v_2) = (n_2 - 1, n_1 - 1)$$

Decision rule

If $|F| \leq F_{(v_1, v_2)}(\alpha)$, **accept** H_0 ; otherwise **reject** H_0 .

Critical region of the F-distribution



10.1 Problem 1 – Two independent samples

Problem 1

Two independent samples of sizes $n_1 = 8$ and $n_2 = 7$ have the following values. Do the estimates of the population variances differ significantly at 5% level of significance?

Sample 1	9	11	13	11	15	9	12	14
Sample 2	10	12	10	14	9	8	10	

Sample variances

$$s_1^2 = \frac{\sum x_{1i}^2}{n_1} - \left(\frac{\sum x_{1i}}{n_1} \right)^2 = \frac{1138}{8} - \left(\frac{94}{8} \right)^2 \Rightarrow \boxed{s_1^2 = 4.1875}$$

$$s_2^2 = \frac{\sum x_{2i}^2}{n_2} - \left(\frac{\sum x_{2i}}{n_2} \right)^2 = \frac{785}{7} - \left(\frac{73}{7} \right)^2 \Rightarrow \boxed{s_2^2 = 3.387}$$

Estimates of the population variances

$$\hat{\sigma}_1^2 = \left(\frac{n_1}{n_1 - 1} \right) s_1^2 = \left(\frac{8}{7} \right) (4.1875) \Rightarrow \boxed{\hat{\sigma}_1^2 = 4.785}$$

$$\hat{\sigma}_2^2 = \left(\frac{n_2}{n_2 - 1} \right) s_2^2 = \left(\frac{7}{6} \right) (3.387) \Rightarrow \boxed{\hat{\sigma}_2^2 = 3.951}$$

Hypotheses

Null Hyp. (H_0): $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$

(No significant difference between the population variances given by the two samples.)

Alt. Hyp. (H_1): $\hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$

(There is some significant difference between the population variances.)

Test statistic (since $\hat{\sigma}_1^2 > \hat{\sigma}_2^2$)

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} = \frac{4.785}{3.951} \Rightarrow \boxed{F = 1.211}$$

Critical value

For $(v_1, v_2) = (n_1 - 1, n_2 - 1) = (7, 6)$ degrees of freedom and $\alpha = 5\% \Rightarrow$

$$\boxed{F_{(7,6)}(0.05) = 4.21} \quad (\text{from the tables})$$

Comparison

$$|F| = 1.211 < 4.21 = F_{(v_1, v_2)}(\alpha) \Rightarrow \text{Accept } H_0$$

Conclusion

There is no significant difference between the population variances given by the two samples.

10.2 Note – Samples from the same normal population**Both tests are needed**

For problems such as “test whether the two samples are taken from the same normal population”, we have to adopt both the t -test and the F -test, to test the significance of the difference between

- the sample mean values, and
- the population variances.

If H_0 is accepted in **both** tests, we conclude that the samples are taken from the same normal population.

10.3 Problem 2 – Nicotine content of two samples**Problem 2**

The nicotine content of two random samples of tobacco are given below. Can you say that the two samples are taken from the same population?

Sample 1	21	24	25	26	27	
Sample 2	22	27	28	30	31	36

We have to perform

1. t -test for the difference between the two sample means.
2. F -test for the difference between the population variances given by the samples.

Solution – Given

$$n_1 = 5, \quad \bar{X}_1 = \frac{21 + 24 + 25 + 26 + 27}{5} \Rightarrow \boxed{\bar{X}_1 = 24.6}$$

$$n_2 = 6, \quad \bar{X}_2 = \frac{22 + 27 + 28 + 30 + 31 + 36}{6} \Rightarrow \boxed{\bar{X}_2 = 29}$$

$$s_1^2 = \frac{\sum x_{1i}^2}{n_1} - (\bar{X}_1)^2 = \boxed{4.24}, \quad s_2^2 = \frac{\sum x_{2i}^2}{n_2} - (\bar{X}_2)^2 = \boxed{18}$$

(i) *t*-test for the difference between the sample means**Hypotheses****Null Hyp. (H_0):** $\bar{X}_1 = \bar{X}_2$

(No significant difference between the two sample means.)

Alt. Hyp. (H_1): $\bar{X}_1 \neq \bar{X}_2$

(There is some significant difference between the two sample means.)

 $\Rightarrow$ **The test is a two tailed test.****Test statistic ($n_1 \neq n_2 \Rightarrow$ Case (i))**

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{24.6 - 29}{\sqrt{\left(\frac{5(4.24) + 6(18)}{9}\right) \left(\frac{1}{5} + \frac{1}{6}\right)}}$$

$$\Rightarrow \boxed{t = -1.918}$$

Critical value and decisionFor a two tailed test, $\alpha = 5\%$ and $v = n_1 + n_2 - 2 = 9$ degrees of freedom,

$$t_9(0.025) = 2.262.$$

$$|t| = 1.918 < 2.262 \Rightarrow \text{Accept } H_0$$

(ii) *F*-test for the population variances**Estimates of the population variances**

$$\hat{\sigma}_1^2 = \frac{n_1}{n_1 - 1} s_1^2 = \frac{5}{4}(4.24) = \boxed{5.3}, \quad \hat{\sigma}_2^2 = \frac{n_2}{n_2 - 1} s_2^2 = \frac{6}{5}(18) = \boxed{21.6}$$

Hypotheses**Null Hyp. (H_0):** $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$

(No significant difference between the population variances.)

Alt. Hyp. (H_1): $\hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$

(There is some significant difference in the population variances.)

Test statistic (since $\hat{\sigma}_2^2 > \hat{\sigma}_1^2$)

$$F = \frac{\hat{\sigma}_2^2}{\hat{\sigma}_1^2} = \frac{21.6}{5.3} \Rightarrow \boxed{F = 4.075}$$

Critical value

For $(v_1, v_2) = (n_2 - 1, n_1 - 1) = (5, 4)$ degrees of freedom and $\alpha = 5\% \Rightarrow$

$$F_{(5,4)}(0.05) = 6.26$$

Comparison

$$|F| = 4.075 < 6.26 = F_{(v_1, v_2)}(\alpha) \Rightarrow \text{Accept } H_0$$

Conclusion

H_0 is accepted in **both** the t -test and the F -test.

Hence the two samples are taken from the same normal population.

11 χ^2 -Test (Chi-squared)

Uses

The χ^2 -distribution is used to test

1. Goodness of fit,
2. Independence of attributes.

Limitations

1. The total number of frequencies N of the distribution must be ≥ 50 .
2. The number of classes n for dividing the frequency must lie between $4 \leq n \leq 16$.
3. The value of each observed frequency $O_i \geq 10$. If it is less than 10, it can be combined with a nearby cell frequency.

11.1 Test of Goodness of Fit

Idea

Through the test of goodness of fit we can identify whether the data meets a certain hypothetical assumption. We have the observed frequencies O_i ($1 \leq i \leq n$) of each class, and based on the hypothetical assumption we construct the expected frequencies E_i ($1 \leq i \leq n$) such that $\sum O_i = \sum E_i$.

Hypotheses

Null Hyp. (H_0): The given data meets the hypothetical assumption (i.e. the given data perfectly fits the condition that we considered.)

Alt. Hyp. (H_1): The given data does not meet that assumption.

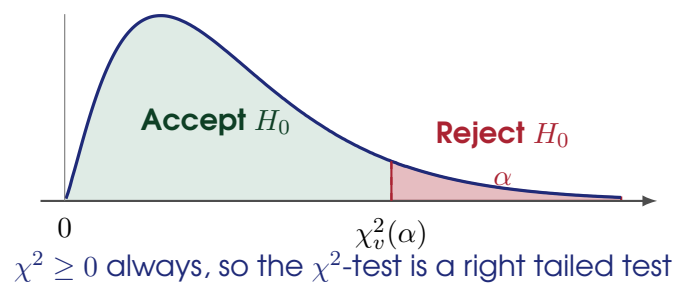
Test statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

with $v = n - 1$ degrees of freedom.

Decision rule

If $\chi^2 < \chi_v^2(\alpha)$, where α is the level of significance, then **accept** H_0 ; otherwise **reject** H_0 .

Critical region of the χ^2 -distribution**11.2 Problem 1 – Performance of four machines****Problem 1**

The following data shows the defective articles produced by 4 machines. Do the data indicate that a significant difference exists in the performance of the machines?

Machines	A	B	C	D
Production time (in hrs)	1	1	2	3
No. of defectives	12	30	63	98

Solution – Given

$n = 4$, $N = 203$, observed frequencies O_i : 12, 30, 63, 98

Hypotheses

Null Hyp. (H_0): There is no significant difference in the performance of the machines.

Alt. Hyp. (H_1): There is a significant difference in the performance of the machines.

Expected frequencies (in the ratio of the production times 1 : 1 : 2 : 3, total 7)

$$E_i : \frac{1}{7} \times 203, \frac{1}{7} \times 203, \frac{2}{7} \times 203, \frac{3}{7} \times 203 \Rightarrow E_i : 29, 29, 58, 87$$

Test statistic

$$\begin{aligned} \chi^2 &= \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(12 - 29)^2}{29} + \frac{(30 - 29)^2}{29} + \frac{(63 - 58)^2}{58} + \frac{(98 - 87)^2}{87} \\ &= 9.9655 + 0.0345 + 0.4310 + 1.3908 \Rightarrow \boxed{\chi^2 = 11.822} \end{aligned}$$

Critical value

$$\text{For } v = n - 1 = 3 \text{ degrees of freedom and } \alpha = 5\% \Rightarrow \boxed{\chi_{0.05}^2(3) = 7.815}$$

Comparison

$$\chi^2 = 11.822 > 7.815 = \chi_v^2(\alpha) \Rightarrow \text{Reject } H_0 \Rightarrow \text{Accept } H_1$$

Conclusion

There is a significant difference in the performance of the machines.

11.3 Problem 2 – Examination results

Problem 2

A sample analysis of exam results of 500 students was made. It was found that 200 failed in the examination, 170 secured third class, 90 obtained second class and 40 secured first class. Do these figures support the belief that the results are in the ratio 4 : 3 : 2 : 1?

Solution – Given

$$n = 4, \quad N = 500, \quad \text{observed frequencies } O_i : 200, 170, 90, 40$$

Hypotheses

Null Hyp. (H_0): The results are in the ratio 4 : 3 : 2 : 1.

Alt. Hyp. (H_1): The results are not in the ratio 4 : 3 : 2 : 1.

Expected frequencies

$$E_i = \frac{4}{10}(500), \frac{3}{10}(500), \frac{2}{10}(500), \frac{1}{10}(500) \Rightarrow E_i : 200, 150, 100, 50$$

Test statistic

$$\chi^2 = \frac{(200 - 200)^2}{200} + \frac{(170 - 150)^2}{150} + \frac{(90 - 100)^2}{100} + \frac{(40 - 50)^2}{50}$$

$$= 0 + 2.6667 + 1 + 2 \Rightarrow \boxed{\chi^2 = 5.67}$$

Critical value

For $v = n - 1 = 3$ degrees of freedom and $\alpha = 5\%$ LOS $\Rightarrow \boxed{\chi_{0.05}^2(3) = 7.815}$

Comparison

$$\chi^2 = 5.67 < 7.815 = \chi_{0.05}^2(3) \Rightarrow \text{Accept } H_0$$

Conclusion

The exam results follow the ratio 4 : 3 : 2 : 1 satisfactorily.

11.4 Problem 3 – Boys and girls distribution**Problem 3**

A survey of 320 families with 5 children each gave the following distribution. Is this result consistent with the hypothesis that male and female births are equally probable?

No. of boys	0	1	2	3	4	5
No. of families	12	40	88	110	60	10

Hypotheses

Null Hyp. (H_0): Male and female births are equally probable, i.e. $X \sim \text{Binomial}(n = 5, p = \frac{1}{2})$, where X is the number of boys in 5 births.

Alt. Hyp. (H_1): Male and female births are not equally probable.

Expected frequencies (with $N = \sum O_i = 320$)

$$P(X = x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x} = \frac{{}^5C_x}{32}, \quad x = 0, 1, 2, 3, 4, 5$$

$$E_i = P(X = i) \times 320 = \frac{{}^5C_i}{32} \times 320 = 10 {}^5C_i$$

$$E(0) = 10, \quad E(1) = 50, \quad E(2) = 100, \quad E(3) = 100, \quad E(4) = 50, \quad E(5) = 10$$

Test statistic

$$\chi^2 = \frac{(12 - 10)^2}{10} + \frac{(40 - 50)^2}{50} + \frac{(88 - 100)^2}{100} + \frac{(110 - 100)^2}{100} + \frac{(60 - 50)^2}{50} + \frac{(10 - 10)^2}{10}$$

$$= 0.4 + 2 + 1.44 + 1 + 2 + 0 \Rightarrow \boxed{\chi^2 = 6.84}$$

Critical value

For $v = n - 1 = 6 - 1 = 5$ degrees of freedom and $\alpha = 5\% \Rightarrow \boxed{\chi_{0.05}^2(5) = 11.07}$

Comparison

$$\chi^2 = 6.84 < 11.07 = \chi_{0.05}^2(5) \Rightarrow \text{Accept } H_0$$

Conclusion

The survey data is consistent with the hypothesis that male and female births are equally probable.

11.5 Problem 4 – Goodness of fit of a Poisson distribution**Problem 4**

The following table gives the number of defects observed in 200 items. Fit a Poisson distribution to the data and test the goodness of fit at 5% level of significance.

No. of defects (x)	0	1	2	3	4
No. of items (O_i)	109	65	22	3	1

Hypotheses

Null Hyp. (H_0): The data follow a Poisson distribution.

Alt. Hyp. (H_1): The data do not follow a Poisson distribution.

The parameter λ is estimated by the sample mean

$$\lambda = \bar{x} = \frac{\sum f_i x_i}{N} = \frac{0(109) + 1(65) + 2(22) + 3(3) + 4(1)}{200} = \frac{122}{200} \Rightarrow \boxed{\lambda = 0.61}$$

Expected frequencies (with $e^{-0.61} = 0.5434$ and $E_i = 200 \times P(X = i)$)

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-0.61} (0.61)^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$E(0) = 200(0.5434) = 108.68, \quad E(1) = E(0) \times \frac{0.61}{1} = 66.30,$$

$$E(2) = E(1) \times \frac{0.61}{2} = 20.22,$$

$$E(3) = E(2) \times \frac{0.61}{3} = 4.11, \quad E(x \geq 4) = 200 - (108.68 + 66.30 + 20.22 + 4.11) = 0.69$$

Here $\sum E_i = 200 = \sum O_i$.

Pooling of classes

The frequencies of the classes $x = 3$ and $x \geq 4$ are less than 10, so they are combined with the class $x = 2$.

Class	O_i	E_i
$x = 0$	109	108.68
$x = 1$	65	66.30
$x \geq 2$	26	25.02
Total	200	200

Number of classes after pooling : $n = 3$.

Test statistic

$$\chi^2 = \frac{(109 - 108.68)^2}{108.68} + \frac{(65 - 66.30)^2}{66.30} + \frac{(26 - 25.02)^2}{25.02}$$

$$= 0.0009 + 0.0255 + 0.0384 \Rightarrow \boxed{\chi^2 = 0.065}$$

Critical value

Since λ was estimated from the data, one extra degree of freedom is lost:

$$v = n - 1 - 1 = 3 - 2 = 1.$$

$$\text{At } \alpha = 5\% \Rightarrow \boxed{\chi_{0.05}^2(1) = 3.841}$$

Comparison

$$\chi^2 = 0.065 < 3.841 = \chi_{0.05}^2(1) \Rightarrow \text{Accept } H_0$$

Conclusion

The Poisson distribution is a good fit to the given data.

12 Exercises

Exercise 1

A sample of 100 students is taken from a large population. The mean height of the students in this sample is 160 cm. Can it be reasonably regarded that this sample is from a population of mean 165 cm and S.D. 10 cm? Also find the 95% fiducial limits for the mean.

Exercise 2

The mean lifetime of a sample of 100 light tubes produced by a company is found to be 1580 hours with a standard deviation of 90 hours. Test the hypothesis that the mean lifetime of the tubes produced by the company is 1600 hours.

Exercise 3

The average hourly wage of a sample of 150 workers in plant *A* was Rs. 2.56 with a standard deviation of Rs. 1.08. The average hourly wage of a sample of 200 workers in plant *B* was Rs. 2.87 with a standard deviation of Rs. 1.28. Can an applicant safely assume that the hourly wages paid by plant *B* are higher than those paid by plant *A*?

Exercise 4

The mean produce of wheat from a sample of 100 fields comes to 200 kg per acre and another sample of 150 fields gives a mean of 220 kg per acre. Assuming the S.D. of the yield at 11 kg for the universe, test if there is a significant difference between the means of the samples.

Exercise 5

In a certain factory there are two independent processes manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 Ozs, with a standard deviation of 12 Ozs, while the corresponding figures in a sample of 400 items from the other process are 124 and 14. Is the difference between the two sample means significant? Also find the 99% confidence limits for the difference in the average weight of the items produced by the two processes.

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